NUST
School of Electrical Engineering & Computer Science

CS-251 —Design and Analysis of Algorithms

Assignment 2 — Project Proposal

Autonomous Robot Path Planning Using Rapidly-Exploring Random Trees (RRT)

---The purpose of this assignment is to develop your ability to:

• Identify and formalize a real-world computational problem

• Analyze its algorithmic challenges

• Conduct a brief literature review

• Select and justify an appropriate algorithm (not covered in this course)

• Demonstrate understanding through design and preliminary pseudocode This proposal will form the foundation of your semester project.

Team Members:

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1. Real-World Problem Description:

1.1 Problem Context

Autonomous navigation is a key requirement in modern intelligent systems like self-driving cars, warehouse robots, delivery drones, and rescue robots. These systems need to move safely from a starting point to a destination while avoiding obstacles in changing environments.

Traditional search methods often struggle in complex spaces where the number of possible configurations becomes very large. Effective motion planning algorithms are crucial for real-time autonomous decision-making.

This problem affects:

- Robotics engineers designing autonomous agents

- Logistics and warehouse automation industries

- Smart transportation systems

- Disaster response and surveillance operations

Effective path planning directly impacts safety, efficiency, and operational costs.

1.2 Problem Formulation:

Let the workspace be represented as:

$$W \subset \mathbb{R}^2 \text{ or } \mathbb{R}^3$$

where the robot operates in a continuous configuration space.

Inputs:

- Start configuration S
- Goal configuration G
- Set of obstacles O
- Configuration space constraints

Output:

- A collision-free path P connecting S to G

Objective:
Find a feasible path such that:

- The robot avoids all obstacles
- The path is computationally feasible to compute
- Planning time supports near real-time operation

Assumptions:

- Obstacles are known beforehand
- Robot motion is continuous
- Collision detection function is available

2.3 Algorithmic Challenges

The motion planning problem introduces several computational challenges:

- High-dimensional continuous search spaces
- Exponential growth of possible configurations
- Real-time planning constraints
- Non-convex obstacle environments
- Uncertainty and noisy sensing data
- Scalability issues for large environments

The problem is algorithmically meaningful because exhaustive search becomes computationally infeasible.

2. Literature Review

Sampling-based motion planning is now the most common way to solve hard navigation problems.
LaValle (1998) came up with Rapidly-Exploring Random Trees (RRT), a probabilistic algorithm that makes it easy to explore large configuration spaces. The algorithm builds a tree step by step by randomly sampling states and expanding into areas that haven't been explored yet.
Karaman and Frazzoli (2011) put forward RRT*, an enhanced version that offers asymptotic optimality while preserving probabilistic completeness.

Other approaches include:

Method	Strengths	Limitations
Grid-based Search	Simple implementation	Poor scalability
Potential Field Methods	Fast computation	Local minima issues
Probabilistic Roadmaps (PRM)	Efficient multi-query planning	Expensive preprocessing
RRT	Fast exploration	Non-optimal paths

Research Gap:
Many deterministic planners struggle with scalability in continuous high-dimensional environments. A sampling-based planner like RRT provides efficient exploration but still requires analysis of trade-offs between optimality and computational efficiency.

References

- LaValle, S. M. (1998). Rapidly-Exploring Random Trees.
- Karaman, S., & Frazzoli, E. (2011). Sampling-based algorithms for optimal motion planning.
- Choset, H. et al. (2005). Principles of Robot Motion.

3. Selected Algorithm :

3.1 Algorithm Selection

The Rapidly-Exploring Random Tree (RRT) algorithm is selected because it is not part of the course syllabus and provides an advanced solution for motion planning in continuous spaces.

3.2 Core Idea

RRT constructs a search tree rooted at the start configuration. At each iteration:

1. Randomly sample a configuration.
2. Find the nearest node in the tree.
3. Extend the tree toward the sampled point.
4. Add the new configuration if collision-free.
5. Repeat until the goal is reached.

The algorithm efficiently explores unexplored regions of the space.

3.3 Complexity Analysis

Let n be the number of nodes generated.

Time Complexity

- Nearest neighbor search: O(n)
- Total expected runtime: O(k·n), where k is iteration count

Space Complexity

- Tree storage: O(n)

3.4 Algorithm Characteristics

Worst Case

- May require many samples before reaching the goal.

Average Case

- Rapid exploration due to probabilistic sampling.

Scalability

- Performs well in high-dimensional spaces compared to deterministic planners.

Trade-offs

- Fast discovery of feasible paths
- Paths are not guaranteed optimal
- May require post-processing smoothing

RRT is probabilistically complete, meaning the probability of finding a solution approaches 1 as iterations increase.

4. Rough Pseudocode

```
Algorithm RRT_PathPlanning(S, G, O):

Initialize tree T with root node S

while Goal not reached do
    q_rand ← RandomSample()
    q_near ← NearestNode(T, q_rand)
    q_new ← Extend(q_near, q_rand)

    if CollisionFree(q_new, O) then
        Add q_new to tree T

Return Path(T, S, G)
```

5. Expected Outcomes

The proposed system will:

- Demonstrate algorithmic understanding of sampling-based planning
- Analyze complexity and scalability behavior
- Provide a foundation for simulation implementation in the semester project
- Explore improvements such as RRT* or path smoothing techniques

6. Final Thoughts

This proposal tackles a genuine computational issue in autonomous navigation by employing a contemporary algorithm not included in the course syllabus. The project focuses on designing algorithms, analyzing their complexity, and making them useful in real life. The Rapidly-Exploring Random Tree algorithm strikes a good balance between theory and practice, making it a good starting point for the semester project.